

CoolRoute Network Planner — Addendum 1

To: PRD v3.0 (Niched, Stress-Tested, Hackathon-Ready)

Date: August 18, 2026

This addendum does not replace the base PRD (`CoolRoute_Network_Planner_PRD_v2.md`). It records decisions made after that document, based on: (1) confirmation that FortyGuard's API coverage is US-only, (2) a detailed endpoint/handbook plan supplied by the team, and (3) a confirmed hackathon credit allotment of 2,000,000. Read this alongside the base PRD — section numbers below reference where each change should be merged in if the documents are combined later.

A1. Scope Validation: US-Only Coverage Does Not Weaken the Problem

FortyGuard's API currently supports data exclusively within the United States. This resolves the earlier open question about international scaling — it's deferred, not abandoned, and the base PRD's §1 "workforce-agnostic" framing still holds *within* the US. The US-only constraint does not weaken the pitch. Supporting evidence:

An estimated 32 million people work outdoors in the US, a population that overlaps heavily with the outdoor mobile workforce persona (couriers, delivery riders, postal, utility, municipal crews).

Extreme heat is the leading cause of weather-related death in the US, killing roughly 2,000 people annually per CDC data — ahead of hurricanes and floods.

High temperatures caused an estimated 28,000 additional workplace injuries in the US in 2023 alone, with injury risk rising sharply as heat index climbs.

Extreme heat's estimated economic cost to the US was around \$162 billion in 2024, with worker-productivity loss as a major component.

Action for the pitch deck/README: replace any "global problem" framing with a tightly-scoped, well-evidenced US framing. A single-country, well-sourced pitch is stronger for a hackathon demo than a vague multi-country one — keep it that way rather than hedging toward "global" language.

Deferred, not cut: international scaling remains architecturally straightforward (see prior discussion — bounding box + local rest-spot data source per country/city) and should be mentioned as explicit future work in the README, contingent on FortyGuard's own geographic expansion.

A2. API Endpoint & Integration Plan (replaces the general description in base PRD §8)

Endpoint	Role in CoolRoute	Priority
<code>POST /v1/heatmap</code>	Core engine. Run on small polygons — route corridors, station buffers, district tiles $\leq$ ~10–20 mi ² . Use historic patterns for baseline (Planner view) and short-range forecast for operations (Dispatcher/Worker views), via <code>filter_type</code> 1 or 3. Granularity set to 80 or 100m to control credit spend.	Highest
<code>POST /v1/env_params</code>	Point queries at existing/proposed cooling stations and key route waypoints for heat index and related metrics. Complements the heatmap grid with accurate point-level data instead of nearest-grid-cell approximation.	High
<code>GET /v1/status/{activity_id}</code>	Async polling with exponential backoff, per the documented pattern. Every <code>heatmap</code> submission requires this — it is not optional.	Required
<code>POST /v1/heat_intelligence</code>	Richer multi-dimensional reports for a small number of key areas, to strengthen the "why" in the Planner narrative and demo. Use only if Premium credits allow.	Medium (bonus)
Satellite / Streetview (Premium)	Land-cover context (greenery vs. asphalt) for what-if scenario realism.	Low / optional

Mapping to product views (new — makes the historic/forecast split explicit):

Planner view → `heatmap` with `filter_type` 1 (historic), since long-horizon infrastructure decisions should rest on stable historic patterns, not a single forecast snapshot.

Dispatcher & Worker views → `heatmap/env_params` with `filter_type` 3 (forecast), since these answer "what should happen in the next few hours," which is inherently operational.

Operating rules (from the FortyGuard handbook, adopted as implementation rules — replaces the general "cache aggressively" language in base PRD §8):

Start with tiny polygons and single timestamps; expand only after the pattern is confirmed working end to end.
Cache every result keyed by (area, date/time) — never re-request an already-cached area/time pair.
Poll politely with exponential backoff on every `status` check.
Stay well under the ~50 mi² / 130 km² AOI limit per request by working at district or corridor scale, never city-wide in a single call.
Use historic data for baseline exposure patterns; use forecast data for operational (Dispatcher/Worker) recommendations. Do not mix the two without labeling which is which in the UI.

A3. AOI Tiling Strategy (new — resolves an AOI-limit gap in base PRD §8)

Phoenix as a whole exceeds the 50 mi² / 130 km² per-request AOI limit by a wide margin, so "cover Phoenix" was never literally achievable in one call — this reinforces, rather than contradicts, the niche-scope decision already made. Concretely:
Define the focus area as a small set of named tiles (e.g., one downtown core + one or two delivery-heavy corridors), each ≤ 10–20 mi², not a single citywide polygon.
The Data Ingestion & Caching Service (base PRD §8) submits one `heatmap` call per tile per required timestamp, polls each independently, and merges results into the local cache.
Any expansion (new tile, new city) is additive — same pattern, more tiles — not a redesign.

A4. Credit Budget: 2,000,000 Credits

FortyGuard does not publicly document a per-call credit cost formula. Rather than estimate blindly, the team should establish the real number empirically before building further:
Day 1 task (add to base PRD §12, Week 1):
Submit one `heatmap` call at the smallest realistic tile size and target granularity (80–100m).
Record the credit consumption reported for that call.
Multiply: (credits per tile) × (number of tiles in the focus area) × (number of timestamps needed — one historic baseline + a few forecast snapshots) × (expected refresh cycles over the two weeks) = projected total spend.
Compare against the 2,000,000 allotment.

Interpretation:

If projected spend is well under budget (a strong likelihood given the tile-based approach and 2M allotment), proceed as planned with headroom for iteration and re-runs.
If projected spend approaches the limit, drop granularity further (e.g., 100m → 150m) or reduce the number of tiles/timestamps before writing any UI code against the data — not after.
This step is now a hard prerequisite for Day 2 (caching-layer build) in the build plan, not an optional check.

A5. Summary of Changes to Merge Into the Base PRD

Base PRD Section	Change
§1 / README framing	Replace "global"/vague framing with the US-specific evidence in A1
§8 (Technical Architecture)	Replace generic caching description with the endpoint table (A2) and tiling strategy (A3)
§11 (Planning Assumptions)	Add: FortyGuard coverage is US-only; international scaling is explicit future work
§12 (Build Plan, Day 1–2)	Insert the credit-budget empirical check (A4) as a hard prerequisite before the caching-layer build
§13 (Risks)	Add: "AOI limit exceeded" and "credit budget unknown" as risks, now mitigated by A3 and A4 respectively